EE 312

EXP 4

RECTIFIERS AND VOLTAGE REGULATORS (AC TO DC CONVERSION)

IN THIS EXPERIMENT:

- Learn how diodes convert ac voltage to unipolar voltage.
- Look at the importance of

✓ filtering

✓ voltage regulation

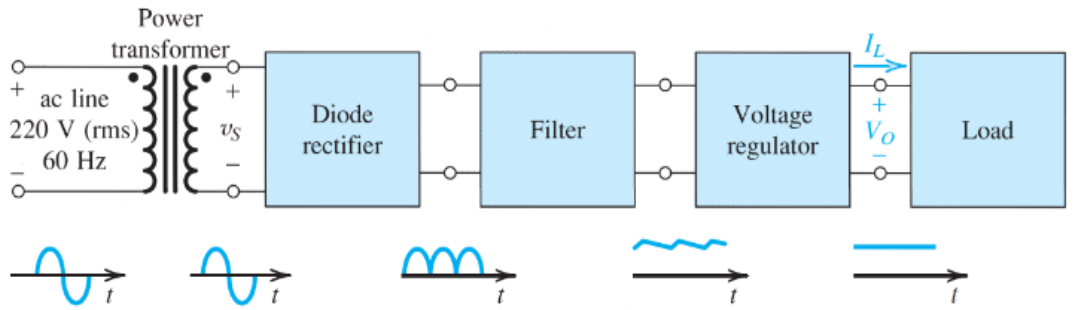
- Learn how to determine maximum ratings for diodes used in rectifier circuits.

Why Do We Need dc Voltage?

- All electronic equipment needs dc voltage to operate.
- The commercially available voltage is ac.
- We need a system to convert from ac to dc .
- This systems takes advantage of the unidirectional current conduction property of diodes.

Block Diagram of a Simple DC Power Supply

- A power transformer steps down the input ac voltage and provides electrical isolation, which is important for safety.
- A rectifier converts ac input voltage to a unipolar pulsating output.
- A filter capacitor reduces the ripple of the resulting output waveform.
- A voltage regulator helps keep the dc voltage fixed despite possible variations in
  - ✓ The current of the load
  - ✓ The input ac voltage



Commercial Power Supplies (AC Adapter)

LINEAR POWER SUPPLY

Low Power Efficiency.

Fixed input ac Voltage.

Low Noise Compared to the Switched type.



SWITCHED POWER SUPPLY

High Power Efficiency.

Accepts a variable input ac voltage.

Suffers from noise.

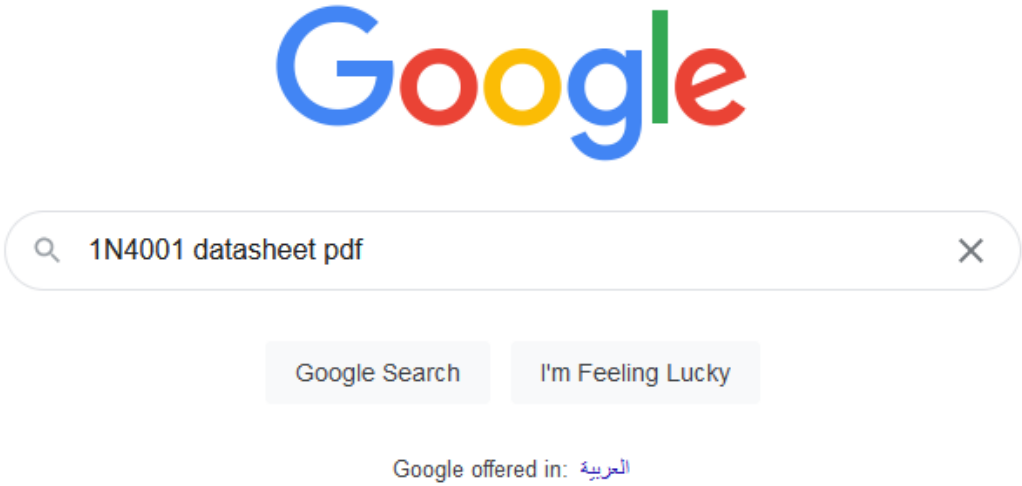


The Rectifier Diode 1N4002



1N4001 Rectifier General Purpose Diode

Searching The Web for a Datasheet

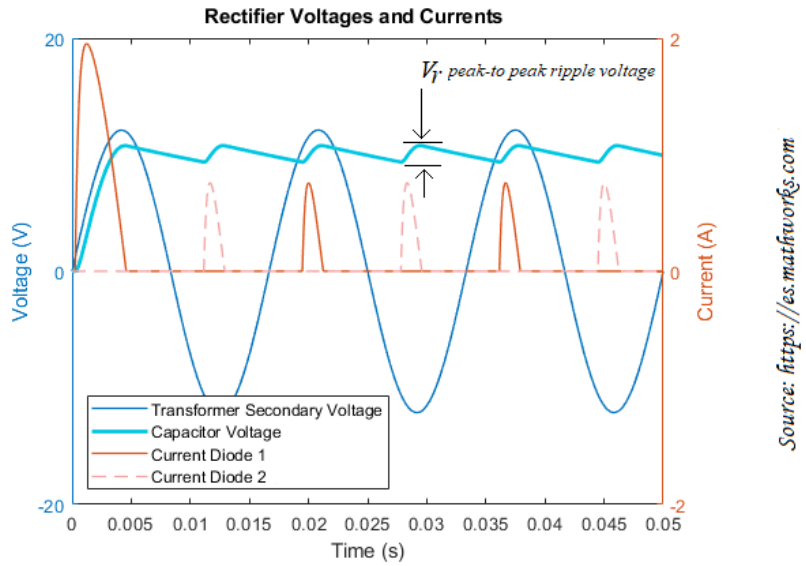


1N400X DATASHEET

V_{RRM} :Repetitive Reverse Voltage (Maximum)

$I_{F(AV)}$: Forward Average Rectified Current

I_{FSM} : Forward Surge Current (Maximum)



Source: https://es.mathworks.com

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

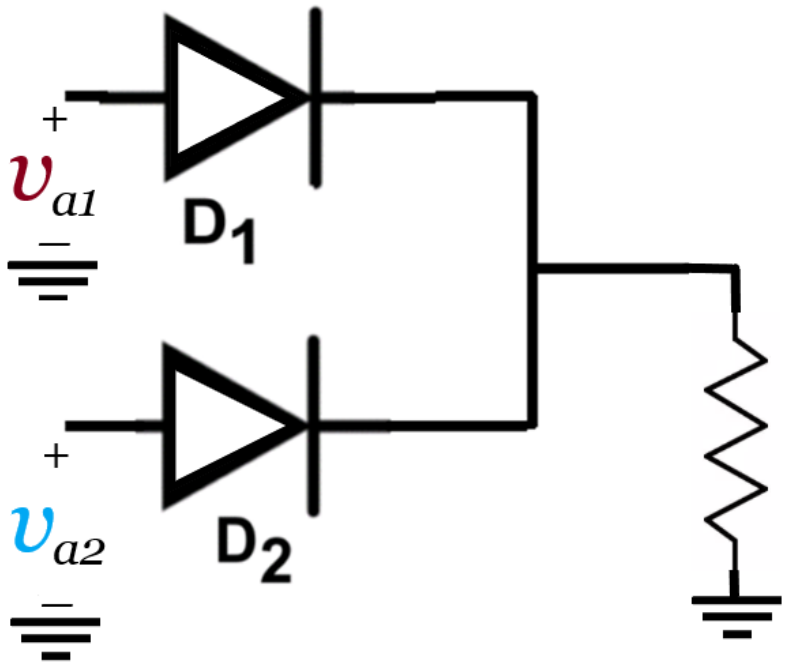
Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

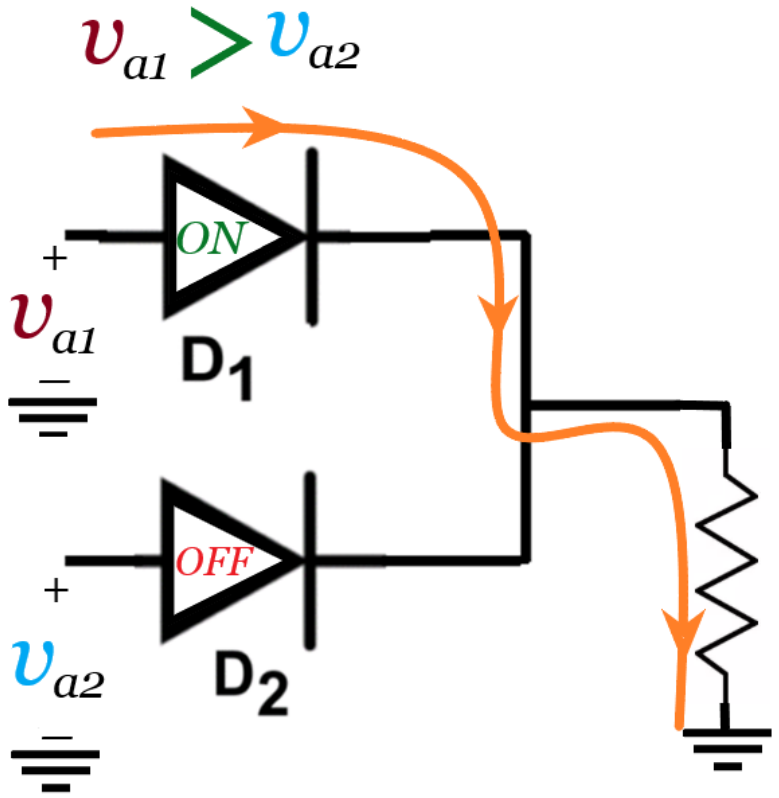
Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Which Diode Conducts Current?

$v_{a1} > v_{a2}$



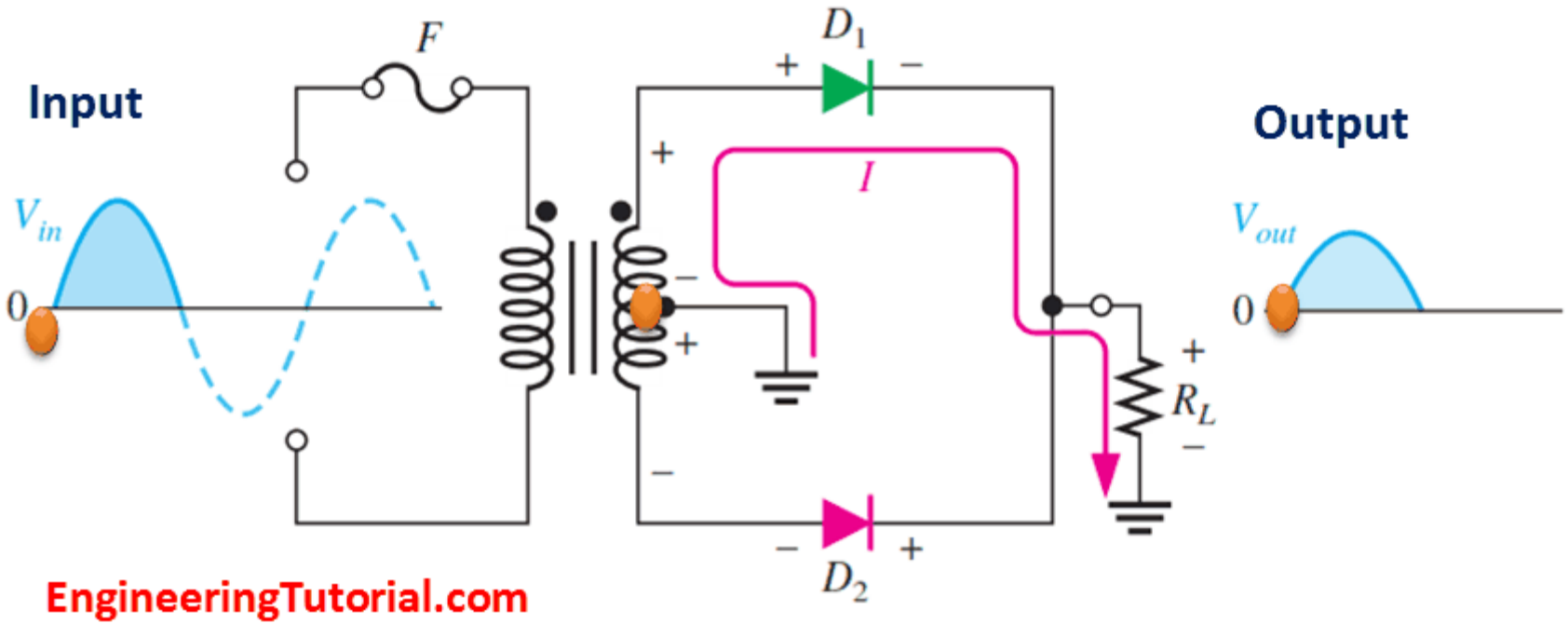
Answer



Center Tapped Full Wave Rectifier

During Positive Half Cycle

D1 : Forward Bias – Closed Circuit



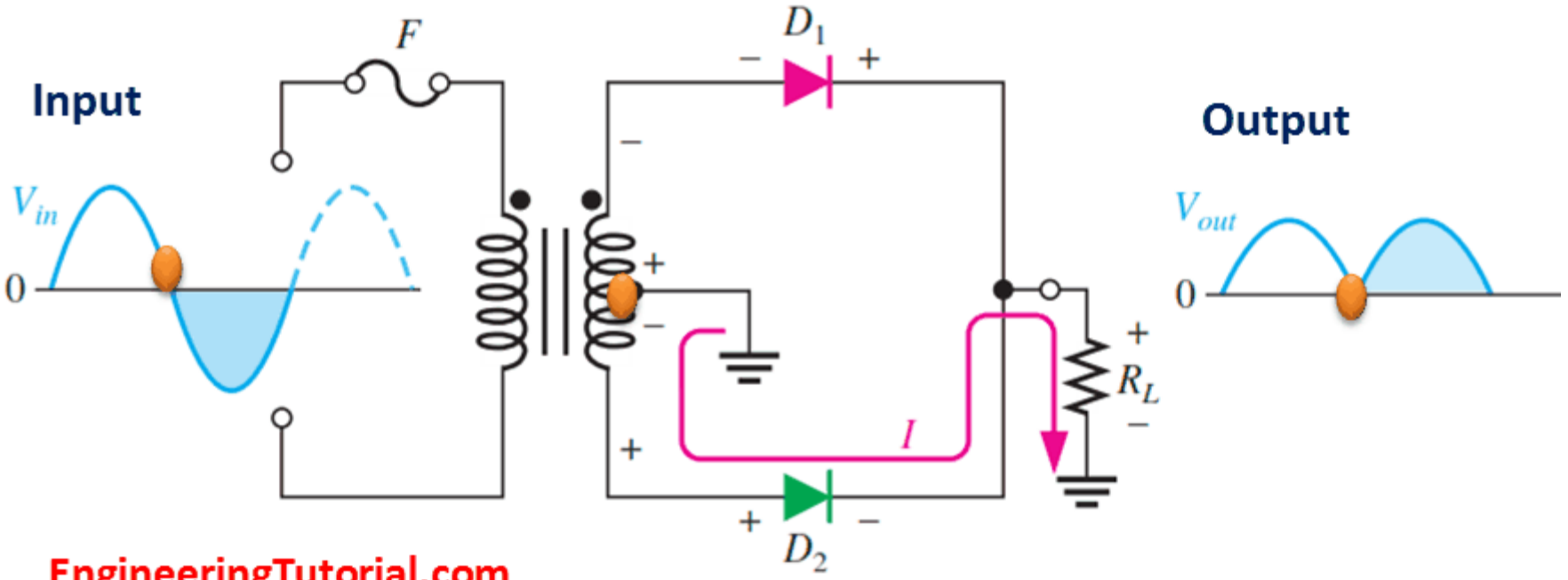
EngineeringTutorial.com

D2 : Reverse Bias – Open Circuit

Center Tapped Full Wave Rectifier

During Negative Half Cycle

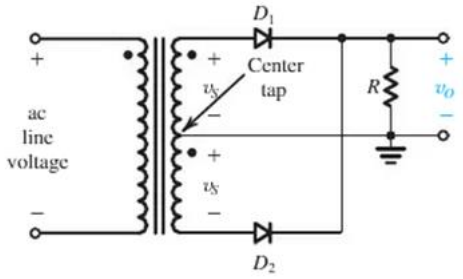
D1 : Reverse Bias – Open Circuit

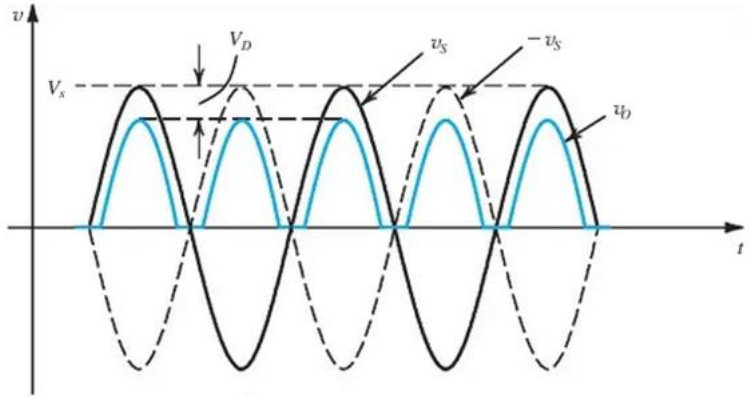


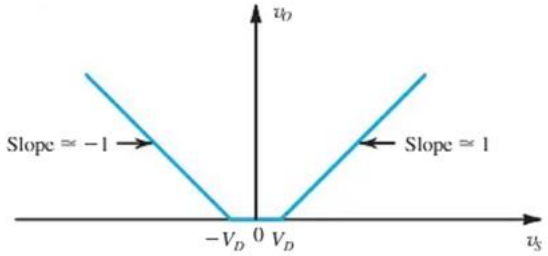
EngineeringTutorial.com

D2 : Forward Bias – Closed Circuit

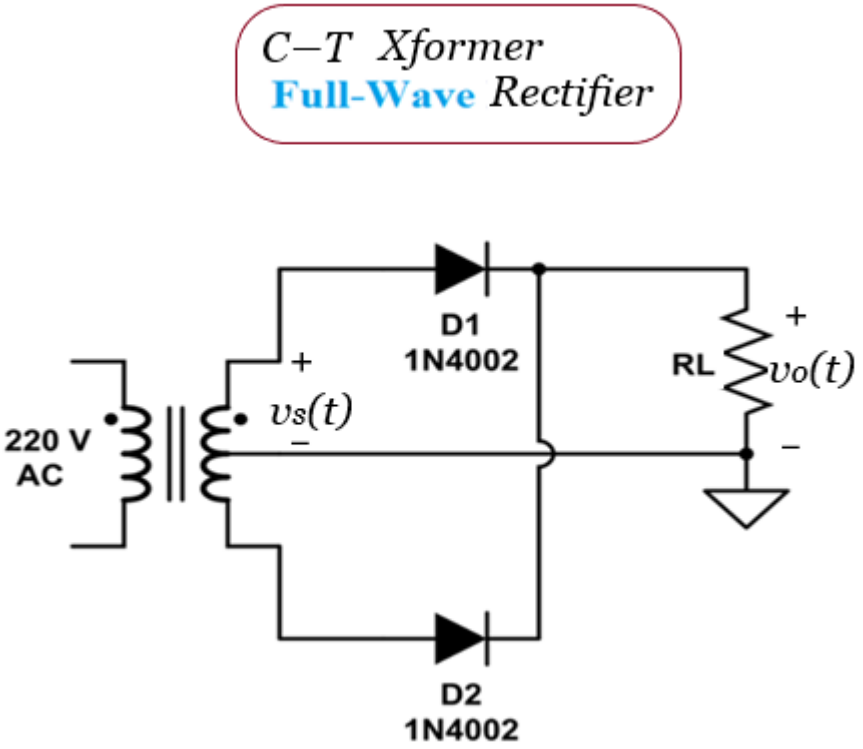
The VTC of Full-Wave Rectifier

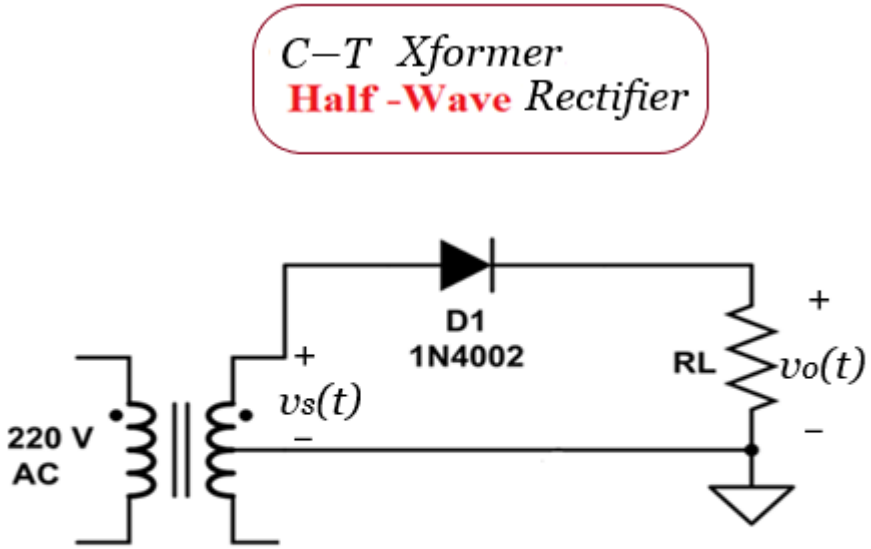






Full-Wave Rectifier vs. Half-Wave Rectifier





CT-Transformer Rectifier PIV Calculation

Buy diodes that have at least

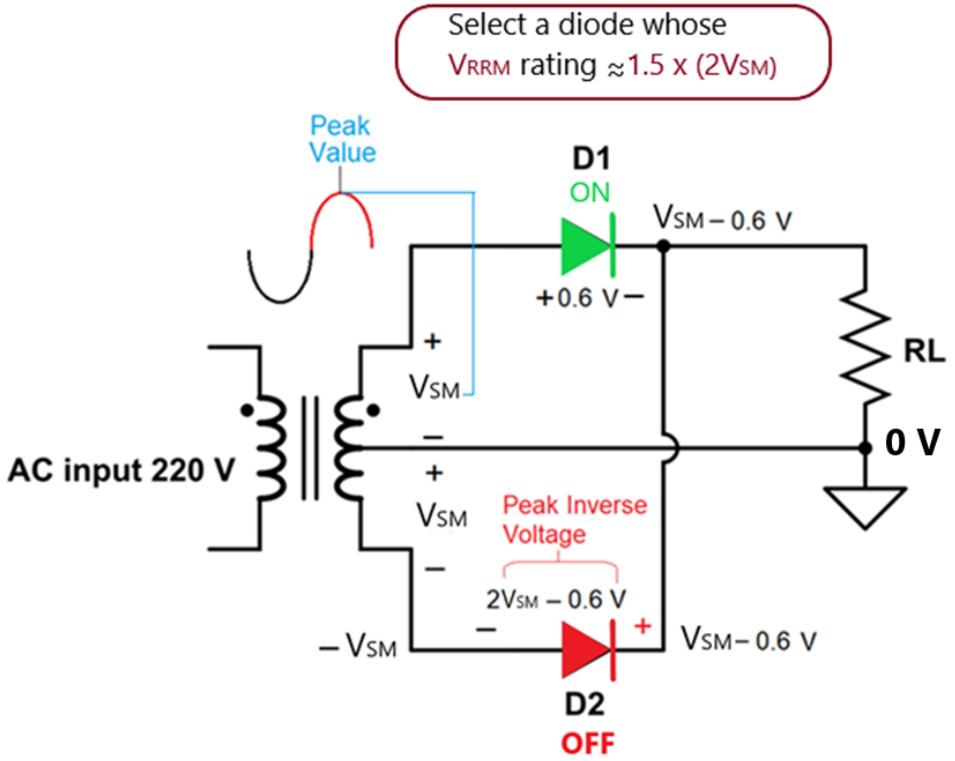
$$V_{RRM} = 1.5 \times PIV \approx 1.5 \times 2\,V_{SM}$$

$$\approx 3\ [\sqrt{2}\ V_s\ (rms)]$$

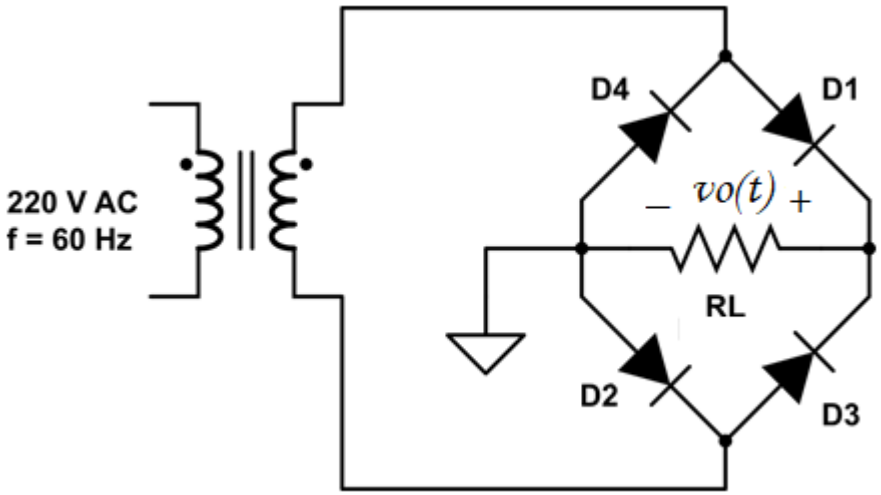
$$V_{RRM} \approx 4\ V_s\ (rms)$$

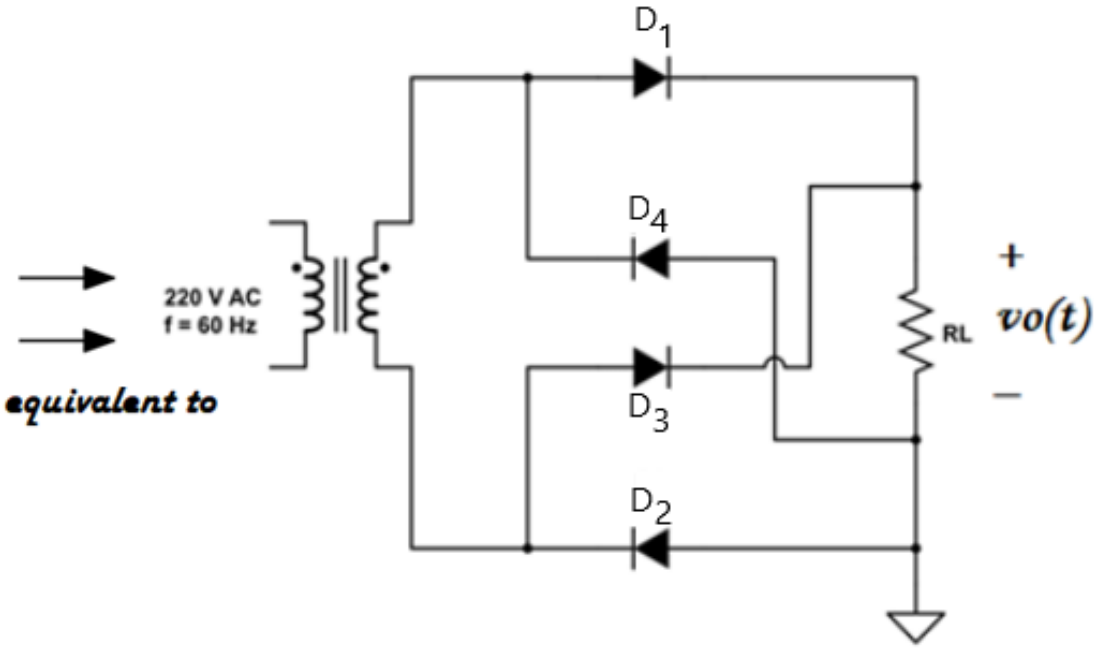
Example: Vs (rms) = 20 V

$V_{RRM} = 4 \times 20 = 80$ V

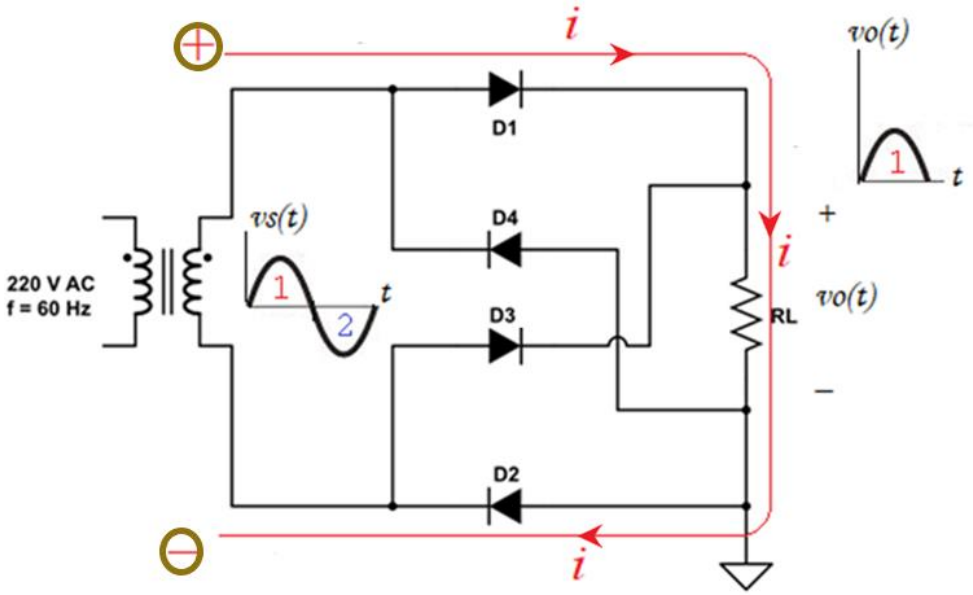


The Bridge Rectifier

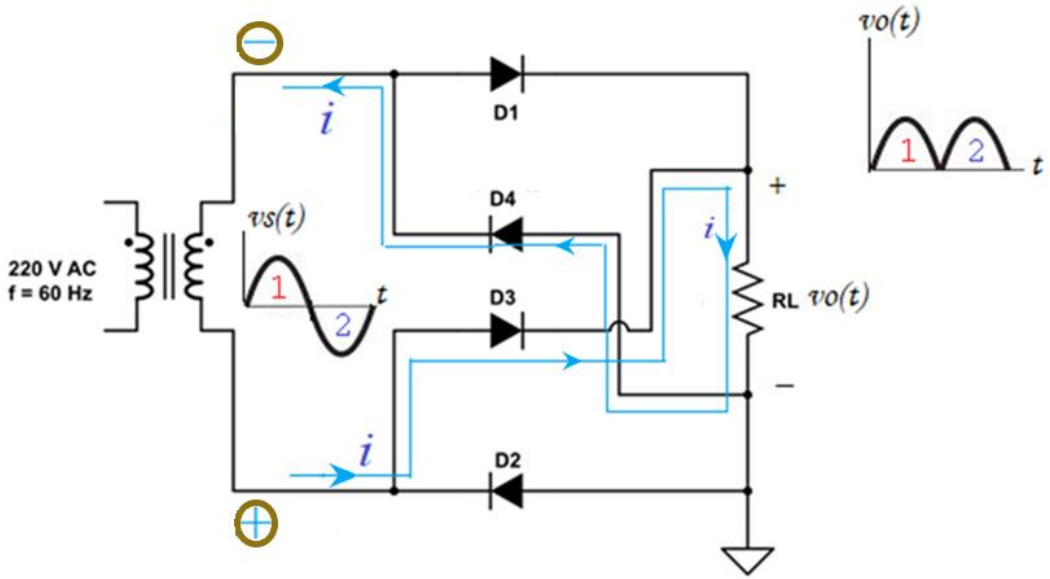




Paths of Current in Bridge Rectifier



Current Path During +ve Half Cycles



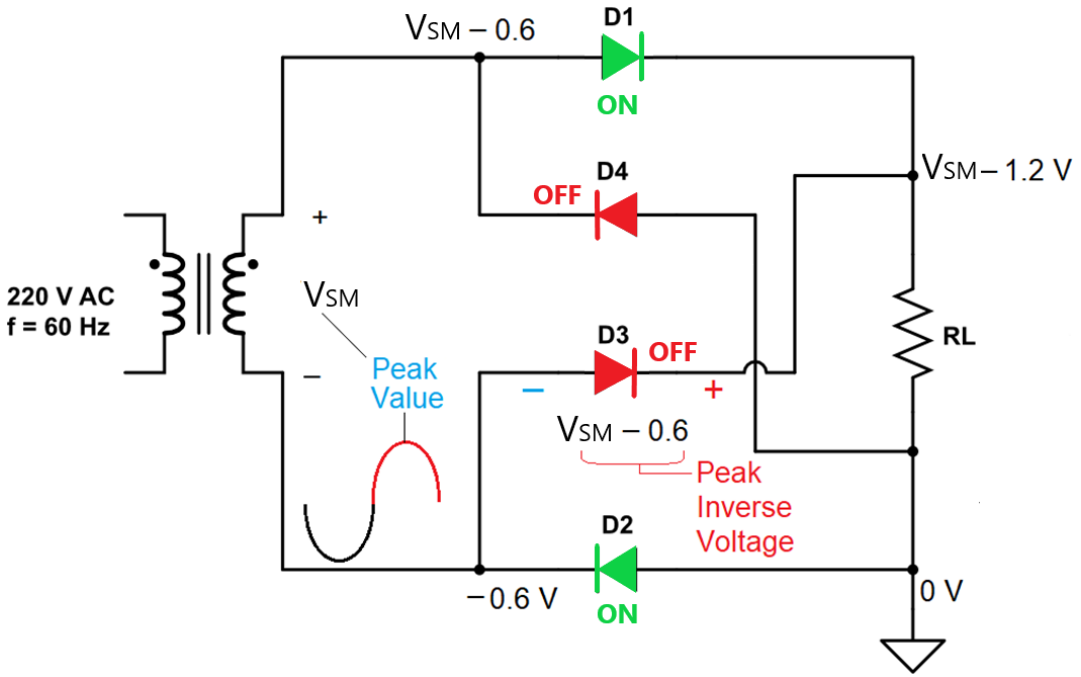
Current Path During -ve Half Cycles

Bridge Rectifier PIV Calculation

Buy diodes that have

$V_{RRM} = 1.5 \times PIV \approx 1.5\ V_{SM}$

$$V_{RRM} \approx 2\ V_s\ (rms)$$



Example: Vs (rms) = 20 V

$V_{RRM} = 2 \times 20 = 40$ V

CT-Xformer Rectifier vs. Bridge Rectifier

	CT-Xformer Rectifier	*Bridge* Rectifier
No. of Diodes	2	4
Cost of Xfromer	High	Low
V_{RRM} for the diodes used	4 Vs (rms)	2 Vs (rms)

In general, it is more economical to use *Bridge* Rectifier

Using a Capacitor to Reduce the Ripple

The peak-to-peak ripple in $vo(t)$ is denoted by $V_{r(pp)}$.

To have best quality dc voltage, we need to reduce $V_{r(pp)}$.

A capacitor is connected across the load to reduce $V_{r(pp)}$.

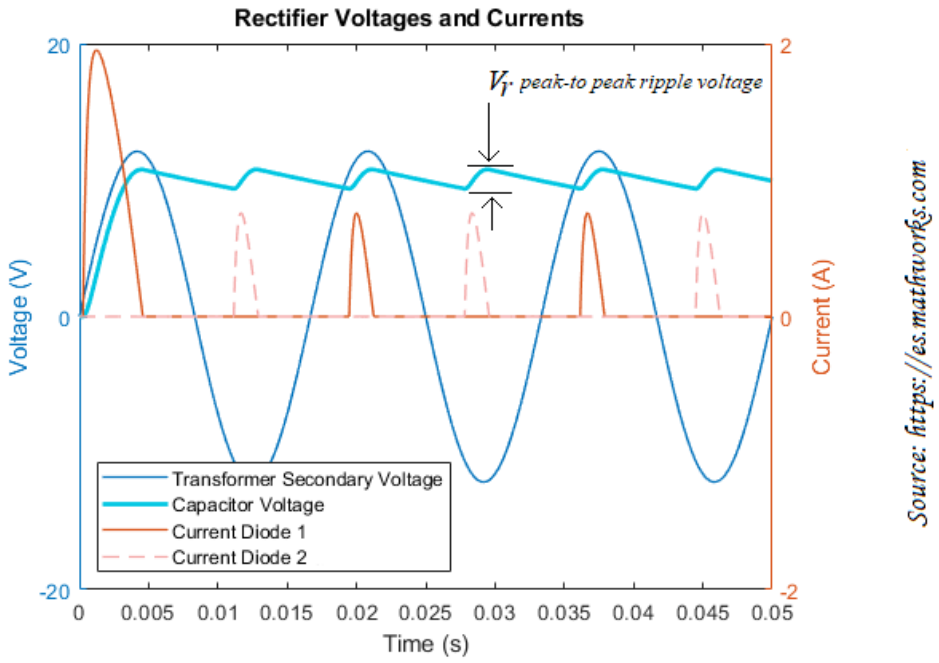
Discharging time is proportional with C and R_L. For a given I_L and V_r, C can be calculated by:

$$C = \frac{1}{120} \times \frac{I_L}{Vr(pp)}$$

The diode average current is given by

$$i_{Dav} \cong I_L \left(1 + \pi \sqrt{\frac{V_{SM}}{2V_{r(pp)}}}\right)$$

Look for a diode with $I_{F(avg)}$ rating = 1.5 i_{Dav}



Source: https://es.mathworks.com

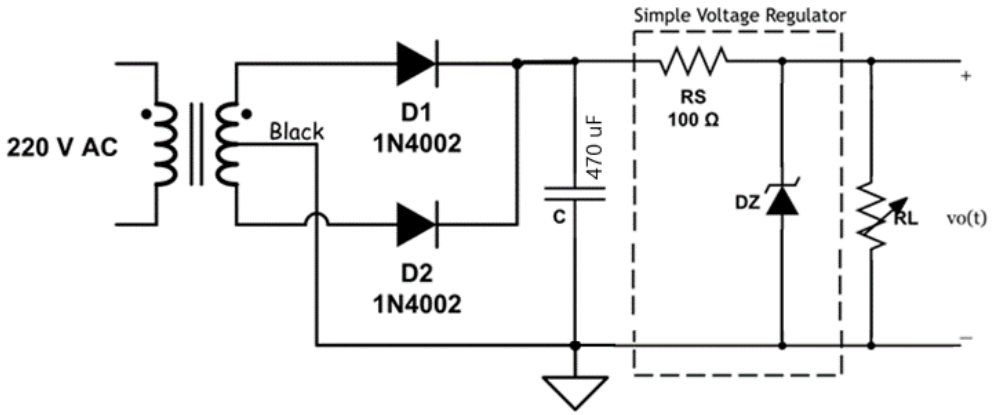
Why Do We Need Voltage Regulator ?

$$V_{r(pp)} = \frac{1}{120} \times \frac{I_L}{C}$$

Note that the ripple voltage is dependent on $R_L \Rightarrow I_L$!

For the same capacitor, more current causes more ripple, which is undesirable.

We need a voltage regulator to make a good-quality output dc voltage for a variable load resistance $R_L \Rightarrow$ variable I_L.

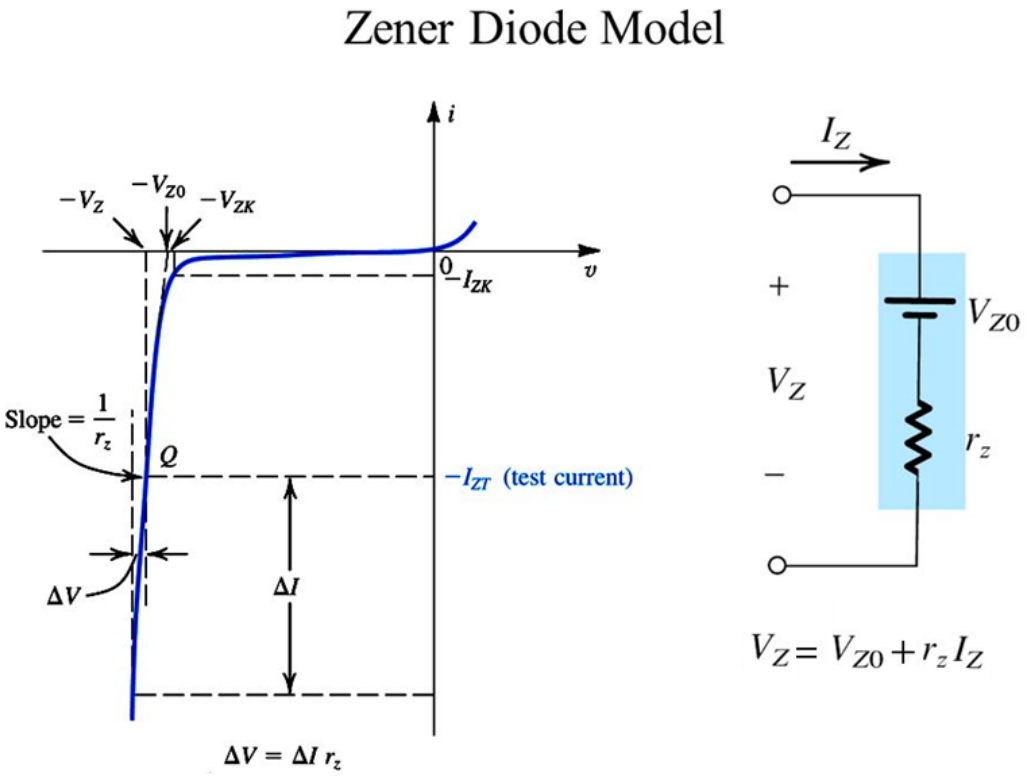


Voltage Regulation Property of the Zener Diode

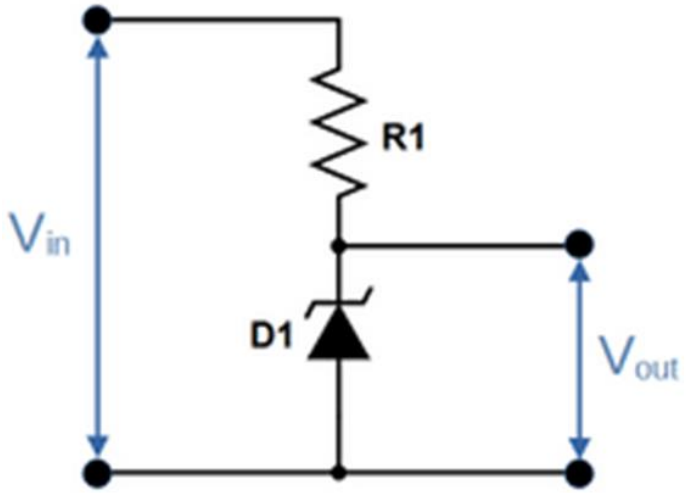
If operated deeply enough in the reverse breakdown region, a Zener diode can serve as a good voltage regulator.

Voltage regulation means *small* ΔV for *large* ΔI. This corresponds to large slope in the *i-v* curve $\Rightarrow$ small r_Z.

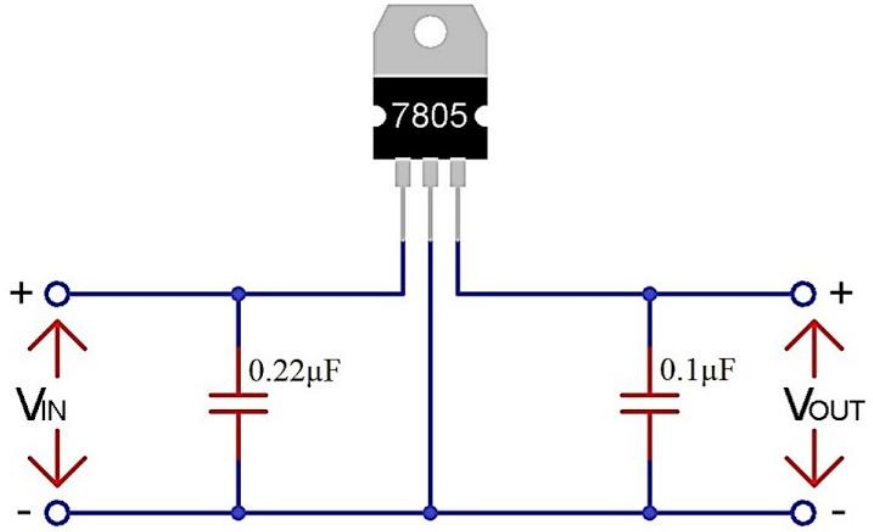
Avoid operating a Zener diode with a current less then the knee current I_{ZK}*, as this will lead to inferior performance.*



Simple Zener Regulator *vs.* Commercial 78XX Regulator



- Less Efficient
- Output voltage usually depends on temperature
- Not suitable for high-power loads.



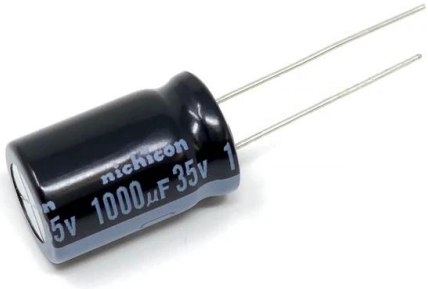
- More Efficient
- Stable with temperature variation
- Available with high power ratings

The Electrolytic Cap Polarity and Max. Voltage

You should connect the long lead to the higher potential node in the circuit, and the shorter lead to the lower potential node. Failure to do so will damage the capacitor.



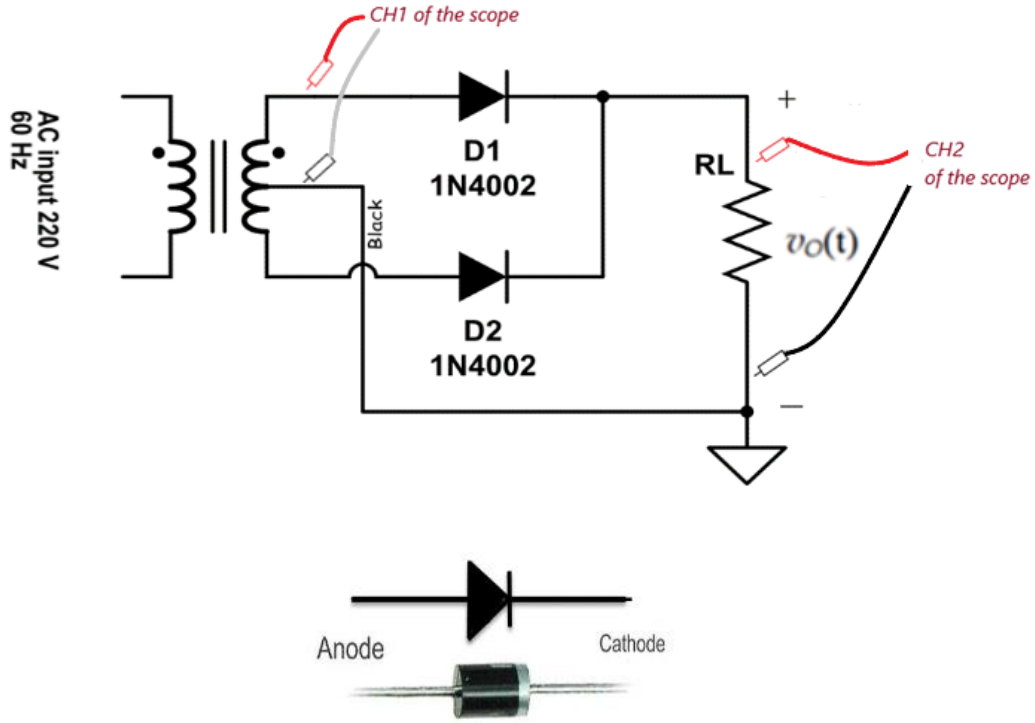
Note that every cap has a maximum operating voltage.

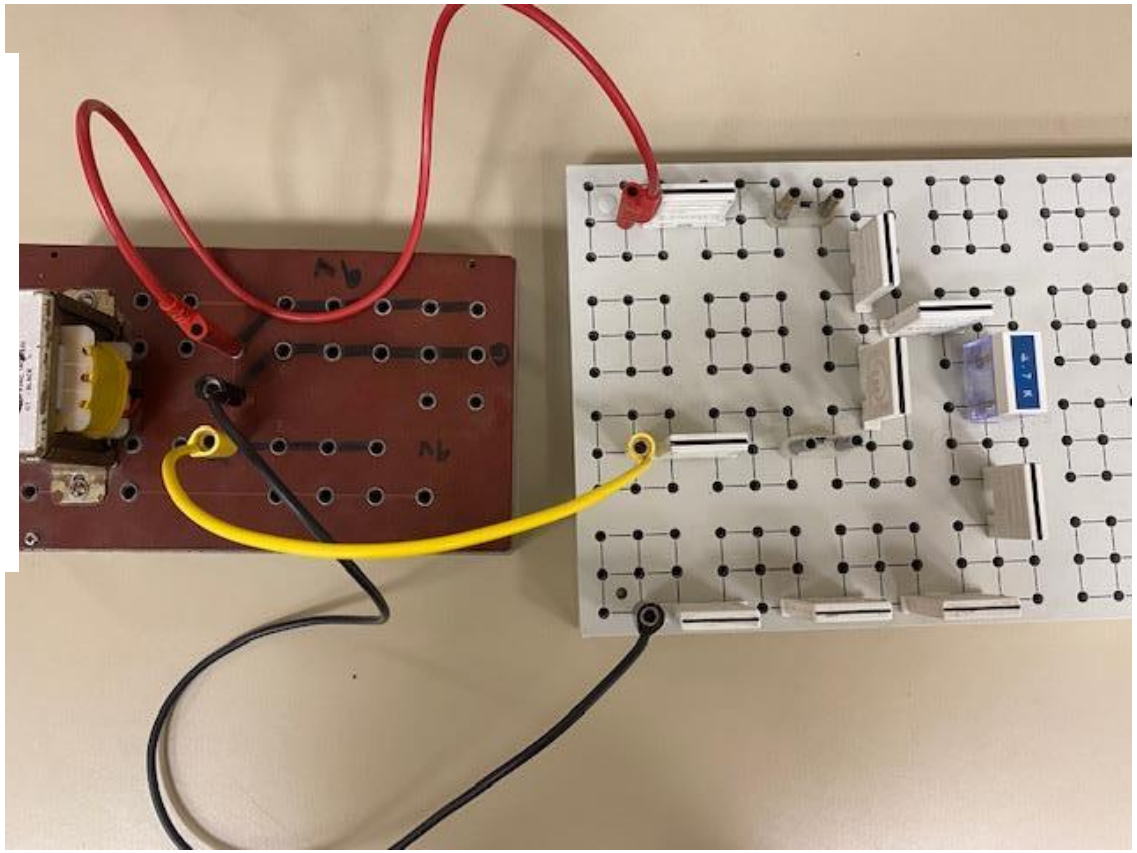


Visually Identifying a Bad Cap Visually



Implementing the CT-Xformer Full Wave Rectifier



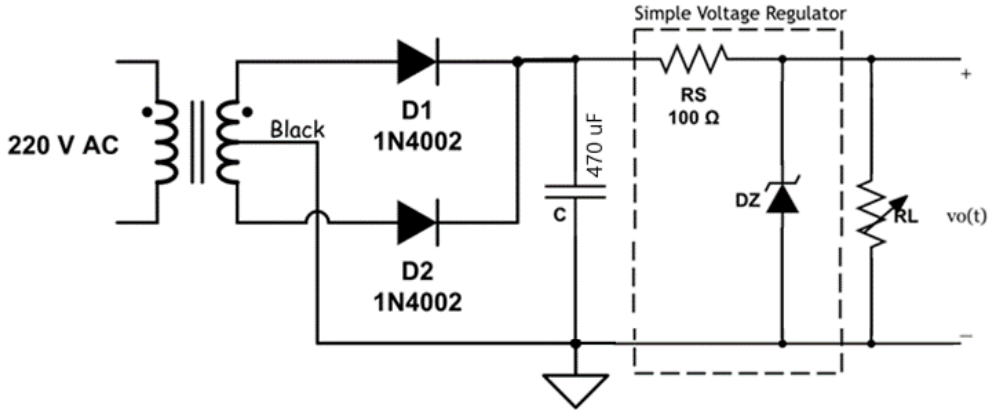


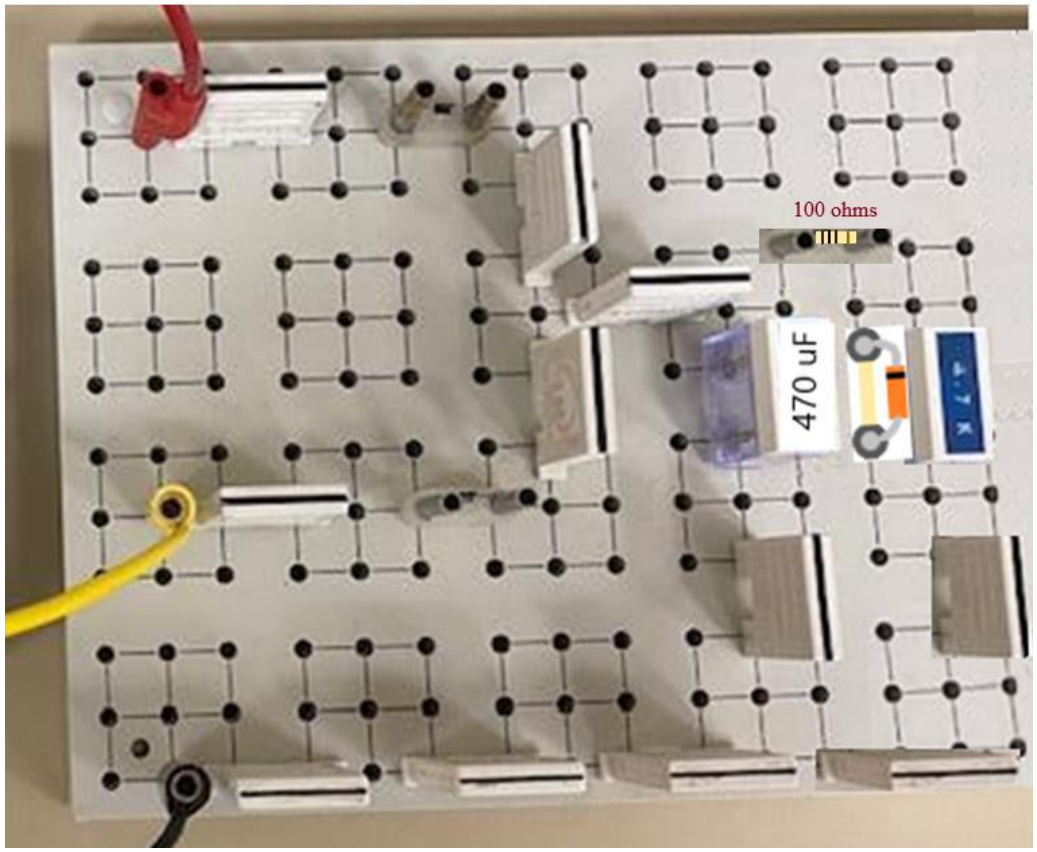
The GwINSTEK GDS-11048 OSCILLOSCOPE

- Press the Default key
- Press Autoset key
- Adjust the Horizontal Scale knob
- Adjust the Vertical Scale
- Adjust the Horizontal Position
- Adjust the Vertical Position
- Add the required Measurements



Setup after Adding Voltage Regulator






LET'S DO IT!

REFERENCES

1. A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, New York: Oxford University Press, 2010.
2. T. A. Almadhi, EE 312 Lab Manual:https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/EE312.pdf